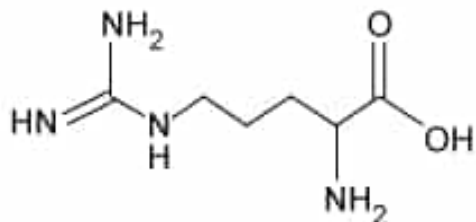


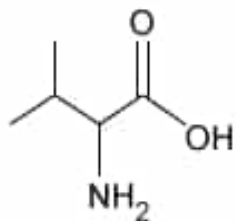
Which of the following amino acids can exist in several diastereomeric forms?

☐ A.



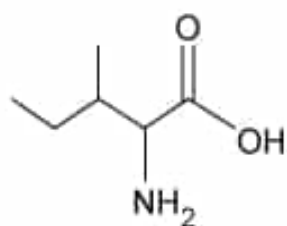
[21%]

☐ B.



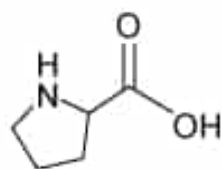
[5%]

☒ C.



[67%]

☐ D.

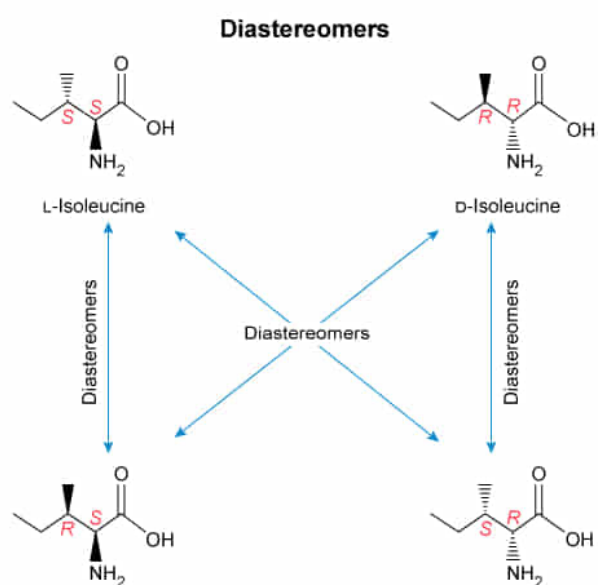


[4%]

Correct

67% answered correctly

Explanation:



One but not all stereocenters have opposite configurations

Stereoisomers are molecules with the same **molecular formula** and atomic connectivity but different spatial arrangements. Stereoisomers contain one or more **stereocenters**, also known as **chiral centers**. **Diastereomers** are stereoisomers that are **not mirror images** of each other and contain **at least two stereocenters**, in which some but not all of the corresponding stereocenters are opposite in configuration.

Amino acids are molecules containing an amine, a carboxyl group, and a side chain, all attached to a central carbon (α -carbon). Because the α -carbon has four different substituents, all amino acids (except for **glycine**) have at least one stereocenter.

Isoleucine (Ile, I) and **threonine (Thr, T)** both have side chains with a stereocenter, giving these amino acids an additional stereocenter for a total of two. Therefore, isoleucine and threonine can exist in several diastereomeric forms.

Only the structure shown in Choice C (isoleucine) has more than one stereocenter. Therefore, of the given options, only the molecule shown in **Choice C** can exist as multiple diastereomers.

(Choices A, B, and D) These structures are arginine (Arg, R), valine (Val, V), and proline (Pro, P), respectively. Each of these amino acids has only *one* stereocenter (α -carbon), and therefore can exist as **enantiomers** but not as diastereomers.

Educational objective:

Diastereomers are stereoisomers that are not mirror images and contain at least two stereocenters. At least one but not all the stereocenters in a pair of diastereomers have opposite configurations.

Time spent: 0 sec

Subject:

Organic Chemistry

QID: 403206

System:

Introduction to Organic Chemistry